

Global Justice Simulator

A webpage that shows users how their income evolves under the **Sustainable Convergence (SC)** scenario from Chancel et al. (2026) and Bothe et al. (2026), part of the Global Justice Report.

Language and tooling

- **Data processing:** R (readxl). Entry point: `code_simulator/prepare_data.R`
- **Webpage:** HTML / CSS / JavaScript (vanilla)
- **Local testing:** XAMPP on Windows 10

Key concepts

Term	Definition
gpercentile	Within-country income percentile group, represented by its lower-bound percentage (0, 1, ..., 98 for standard groups; 99, 99.1, ..., 99.999 for fine top-1% groups)
SC scenario	Sustainable Convergence: global policies reduce inequality to $k=5$ ($T10/B10 \approx 5$) by 2100; world average income converges to ~60,000 EUR PPP 2025
GIT	Global Income Transfers — international lump-sum per-capita redistribution mechanism in the SC scenario
posttax income	After domestic taxes and transfers; "income" in this project also adds GIT (post-GIT)
EUR PPP 2025	All monetary values in constant 2025 purchasing-power-parity euros
k value	Inequality parameter: $k = T10/B10$ ratio; $k=5$ is the SC 2100 target

Data sources

Fetches 2026-05-24:

- **Chancel et al. (2026)** — macro scenarios:
`data/Chancel/Chanceletal2026Appendix_MacroScenarios.xlsx`
- **Bothe et al. (2026)** — within-country distributions:
`data/Bothe/Botheetal2026AppendixDistribution.xlsx`

Key sheets in `Botheetal2026AppendixDistribution.xlsx`

Sheet	Content
P1e	2025 average posttax income by gpercentile group (127 groups × 67 countries). Values in EUR PPP 2025/year. Used as starting distribution.
I2i	SC scenario: average posttax net income time series 1800–2100 (years × 67 countries).

Sheet	Content
I1a	SC scenario: T10 income share time series.
I1b	SC scenario: B50 income share time series.
I1c	SC scenario: T1 income share time series.
I1e	SC scenario: B10 income share time series.
I1h	SC scenario: M40 income share time series.
I5e	SC scenario: P10 income threshold as ratio to average income (pre-GIT, time series).
I5b	SC scenario: P50 income threshold as ratio to average income (pre-GIT, time series).
I5c	SC scenario: P99 income threshold as ratio to average income (pre-GIT, time series).
I5d	SC scenario: P99.9 threshold as ratio to average income (pre-GIT).
I5f	SC scenario: P99.99 threshold as ratio to average income (pre-GIT).
I5g	SC scenario: P99.999 threshold as ratio to average income (pre-GIT).
I8c	SC scenario: T1 income share (post-GIT, time series).
I8e	SC scenario: B10 income share (post-GIT, time series).
I9i	SC scenario: average posttax net income (post-GIT, time series).
K1a/K2a	SC scenario: T10 wealth share / average (time series).
K1b/K2b	B50 wealth share / average.
K1c/K2c	T1 wealth share / average.
K1e/K2e	B10 wealth share / average.
K1h/K2h	M40 wealth share / average.
H1	SC 2100 target distribution shape (k=5): B50=37.7%, M40=44.1%, T10=18.1%, T1=2.6%.
P1a/P1b	Target distribution shapes at various k values (k=1..100).
H3b	Global 2025 income and wealth distribution by global percentile (for reference).

I-series sheets **I8c**, **I8e**, **I9i** (post-GIT) are used as anchors for income.csv. I10 contains post-GIT inequality ratios (T10/B50 etc.), not threshold ratios, so I5 (pre-GIT threshold ratios) is reused as an approximation for the intermediate anchors.

Generated data files

[data/income_pre_git.csv](#)

Posttax income thresholds at 6 key percentiles, **before** Global Income Transfers, for each country and each year 2020–2100.

Column	Description
country	Country/region code (66 values incl. "World")
year	Year, 2020–2100
p10	P10 threshold = I5e × I2i (EUR PPP 2025/year)
p50	P50 threshold = I5b × I2i
p99	P99 threshold = I5c × I2i
p99.9	P99.9 threshold = I5d × I2i
p99.99	P99.99 threshold = I5f × I2i
p99.999	P99.999 threshold = I5g × I2i

Dimensions: 66 countries × 81 years = 5,346 rows.

data/income.csv

SC scenario posttax **post-GIT** income, all 127 gpercentile groups, all years 2020–2100.

Column	Description
country	Country/region code
gpercentile	Lower bound (%) of the percentile bracket: 0, 1, ..., 98 (standard groups, 1 pp wide), then 99, 99.1, ..., 99.8 (0.1 pp), 99.9, 99.91, ..., 99.98 (0.01 pp), 99.99, ..., 99.999 (0.001 pp)
income_2020 ... income_2100	Income in EUR PPP 2025/year for each year

Methodology:

1. **2025 base:** all 127 groups taken directly from P1e (EUR PPP 2025).
2. **Other years:** for each year $t \neq 2025$, compute a **scale factor** at 5 anchor rows as the ratio of the anchor's value in year t to its 2025 value. Log-interpolate between bracketing anchors; multiply by the P1e base value at each row.

Anchor row	Lower bound	Reference value	Series used
1	0	B10 avg = I9i × I8e / 0.1	I9i (post-GIT avg), I8e (post-GIT B10 share)
10	9	P10 threshold = I5e × I9i	I5e (pre-GIT P10 ratio), I9i
50	49	P50 threshold = I5b × I9i	I5b (pre-GIT P50 ratio), I9i

Anchor row	Lower bound	Reference value	Series used
99	98	$P99 \text{ threshold} = I5c \times I9i$	$I5c$ (pre-GIT P99 ratio), $I9i$
100–127	99–99.999	$T1 \text{ avg} = I9i \times I8c / 0.01$	$I9i$ (post-GIT avg), $I8c$ (post-GIT T1 share)

Rows 2–10 are log-interpolated (implicitly assuming a reciprocal income distribution) between anchors 1 and 10; rows 11–50 between 10 and 50; rows 51–99 between 50 and 99; rows 100–127 between 99 and the T1 anchor by their position in [99, 100].

$I9i$ is the post-GIT average from Bothe et al. and is used directly — no manual GIT computation is needed. The intermediate threshold anchors (rows 10, 50, 99) use pre-GIT $I5$ ratios scaled by the post-GIT average $I9i$, since no post-GIT threshold ratio series is available ($I10$ contains inequality ratios, not threshold ratios).

Dimensions: 66 countries $\times$ 127 groups = 8,382 rows $\times$ 83 columns.

data/income_world.csv

World income distribution under SC at 6 target years (2025, 2030, 2035, 2050, 2080, 2100).

Column	Description
<code>gpercentile</code>	Global income percentile rank, 1–100
<code>income_2025 ... income_2100</code>	Income in EUR PPP 2025/year

Dimensions: 100 rows $\times$ 7 columns. Used for the distribution chart on the webpage.

data/wealth.csv

SC scenario wealth, all 100 gpercentiles, target years 2025–2100.

Column	Description
<code>country</code>	Country/region code
<code>gpercentile</code>	Within-country wealth percentile rank, 1–100
<code>wealth_2025 ... wealth_2100</code>	Wealth in EUR PPP 2025

Methodology: 5-group scaling (B10, p10–50, M40, T10–T1, T1) using K2 group average series, starting from K2 group averages at 2025.

Dimensions: 6,600 rows $\times$ 8 columns.

data/wealth_world.csv

World wealth distribution at 6 target years.

Dimensions: 100 rows × 7 columns.

Countries and regions

66 values total, including "World":

Individual countries: AE, AR, AU, BD, BR, CA, CD, CI, CL, CN, CO, DE, DK, DZ, EG, ES, ET, FR, GB, ID, IN, IR, IT, JP, KE, KR, MA, ML, MM, MX, NE, NG, NL, NO, NZ, PH, PK, RU, RW, SA, SD, SE, TH, TR, TW, US, VN, ZA

Residual "other" groups (WID.world codes): OA (Other Russia/Central Asia), OB (Other East Asia), OC (Other Western Europe), OD (Other Latin America), OE (Other MENA), OH (Other North America/Oceania), OI (Other South & SE Asia), OJ (Other Sub-Saharan Africa), QM (Eastern Europe)

Aggregate regions: East Asia, Europe, Latin America, Middle East North Africa, North America Oceania, Russia Central Asia, South & South-East Asia, Sub-Saharan Africa

Plus: World

Running the data pipeline

```
# in R, from the project root:  
Rscript --vanilla code_simulator/prepare_data.R # run from project root
```

Outputs all five CSV files to **data/**. Runtime: ~2 minutes.

Local testing

Place the project folder under XAMPP's **htdocs/** and access via **http://localhost/global_justice/**.